GLM-5.2 - Overview - Z.AI DEVELOPER DOCUMENT

Source: https://docs.z.ai/guides/llm/glm-5.2

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Overview
GLM-5.2 is a flagship model built for the era of long-horizon tasks. With truly usable 1M-token context, it has been tested to handle project-scale engineering context, delivering more stable long-task execution, more reliable adherence to engineering standards, and higher success rates in development scenarios. A single task can complete the full development workflow—from requirements to deployable products across multiple platforms.
Positioning
Flagship Foundation Model
Input Modalities
Text
Output Modalitie
Text
Context Length
1M
Maximum Output Tokens
128K

Capability
Thinking Mode
Offering multiple thinking modes for different scenarios
Streaming Output
Support real-time streaming responses to enhance user interaction experience
Function Call

Powerful tool invocation capabilities, enabling integration with various external toolsets
Context Caching
Intelligent caching mechanism to optimize performance in long conversations
Structured Output
Support for structured output formats like JSON, facilitating system integration
MCP
Flexibly integrate external MCP tools and data sources to expand application scenarios

Usage

Project-Level Codebase Takeover: Let the Model Understand an Entire Project in One Go

Long-Horizon Refactoring: Let It Run a Real Engineering Task End to End

Production-Grade Standards Stress Test: See Whether It Can Hold the Line on Hard Engineering Constraints

Mobile On-Device Debugging Loop: From Code Implementation to Device Validation

WeChat Mini Program Development: Migrating from a Web App to a WeChat Mini Program

Mini Game Development: From Gameplay Rules to a Playable Loop

Research Reproduction: From Paper and Data to a Runnable Engineering Project

Code-to-Video Loop: From Natural-Language Ideas to a Demo-Ready Video

Introducing GLM-5.2
1
1M Context: Making Long-Horizon Tasks Stable and Practical
The foundation of long-horizon tasks is not having a 1M context, but making 1M context truly usable. GLM-5.2 delivers a Solid 1M lossless context and has undergone months of specialized training for long-horizon Coding Agent scenarios, covering high-value tasks such as large-scale implementation, automated research, and performance optimization.
Compared to solutions that merely extend context length, GLM-5.2 maintains more stable performance at ultra-long context, even surpassing Opus in select real-world benchmarks.
GLM-5.2 delivers state-of-the-art long-horizon coding performance among open-source models. Across FrontierSWE, PostTrainBench, and SWE-Marathon, it consistently ranks among the top models overall—trailing Opus 4.8 by just 1% on FrontierSWE, outperforming GPT-5.5 and Opus 4.7 on multiple benchmarks, and remaining the highest-ranked open-source model across all three. These results demonstrate that GLM-5.2's 1M context window translates into practical long-horizon engineering capability.
2
Coding Capabilities Validated by Both Benchmarks and Developers
On standard coding benchmarks, GLM-5.2 is the strongest open-source model, improving on GLM-5.1 by a wide margin: 81.0 vs. 62.0 on Terminal-Bench 2.1 and 62.1 vs. 58.4 on SWE-bench Pro. It also closes much of the gap to the closed-source frontier — on Terminal-Bench 2.1 (81.0) it lands within a few points of Claude Opus 4.8 (85.0) — while staying ahead of Gemini 3.1 Pro.
Before its official release, GLM-5.2 was made available in advance to GLM Coding Plan users. Developers reported improvements mainly in the following areas:
Stronger project-level context capacity, enabling an entire codebase to be placed within a single reasoning workflow;
More stable long-horizon task execution, allowing complex tasks to progress continuously

without easily going off track;
More reliable adherence to production-grade engineering standards, helping enforce hard constraints in team development workflows;
Stronger client-side and mobile engineering capabilities, going beyond app generation to support a complete on-device debugging loop.
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Resources
API Documentation: Learn how to call the API.

Quick Start
The following is a full sample code to help you onboard GLM-5.2 with ease.
cURL
Official Python SDK
Official Java SDK
OpenAI Python SDK
Basic Call
```
curl -X POST "https://api.z.ai/api/paas/v4/chat/completions" \
-H "Content-Type: application/json" \
-H "Authorization: Bearer your-api-key" \
-d '{
  "model": "glm-5.2",
  "messages": [
    {
      "role": "system",
      "content": "You are a senior full-stack software engineer, proficient in frontend development, backend architecture design, and modern web technology stacks."
    },
    {
      "role": "user",
      "content": "Design and build a personal blog website for me, including a homepage, article list page, and article detail page, using React + Node.js technology stack."
    }
  ],
  "thinking": {
    "type": "enabled"
  },
  "reasoning_effort": "max",
  "max_tokens": 4096,
  "temperature": 1.0
}'
```

Streaming Call
```
curl -X POST "https://api.z.ai/api/paas/v4/chat/completions" \
-H "Content-Type: application/json" \
-H "Authorization: Bearer your-api-key" \
-d '{
  "model": "glm-5.2",
  "messages": [
    {
      "role": "system",
      "content": "You are a senior full-stack software engineer, proficient in frontend development, backend architecture design, and modern web technology stacks."
    },
    {
```

```
      "role": "user",
      "content": "Design and build a personal blog website for me, including a homepage, article list page, and article detail page, using React + Node.js technology stack."
    }
  ],
  "thinking": {
    "type": "enabled"
  },
  "reasoning_effort": "max",
  "stream": true,
  "max_tokens": 4096,
  "temperature": 1.0
}'
```


Migrate to GLM-5.2
GLM-5.1